

Manual

Workflows in Escalation Management SIS-HWM 4.0_{pe}

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Contents

1	Overview: Workflows in Escalation Management	4
2	Workflow Overview	5
3	Workflow Events	8
4	Workflow Management - Task List.....	12
5	Workflow History	16
6	Workflow Designer	18
6.1	Starting the Workflow Designer	19
6.2	Workflows	20
6.3	Elements of a Workflow Design.....	24
6.4	Definition of the Working List of a Process Step	25
6.5	Deploying Workflows.....	34
6.6	Import/Export of Workflows	35
6.7	Language / Multilingual input.....	36
6.8	Naming Conventions	37
6.9	Documentation of Created Workflows	38
7	Workflow Management Configuration	40

1 Overview: Workflows in Escalation Management

Purpose

This component allows for triggering workflows on the basis of triggered escalations. Separate workflows can be initiated depending on the type of escalations and the basic information included.

Features

The following functions are available:

- Various workflows according to the type and contents of escalations.
- Automatic initiation of workflows upon the occurrence of an escalation in HYDRA.
- Active information, e.g. by e-mail, upon the activation of workflow steps.
- Automatic or manual activation of workflow steps.
- Assignment and triggering of tasks within a workflow step with deadline allocation and active warning of imminently exceeded deadlines.
- Processing and completion of tasks with possible automatic activation of the next process step.
- Activation of workflow history (which process step/task was completed at what time).

2 Workflow Overview

Overview

Menu	System administration – Workflow management – Workflow overview
Transaction code	wfov
Function authorization	wfov

Usage

The workflow overview provides an overview of currently running, finished and instantiated workflows including status information on individual workflow steps.

Please note:

In order for the user to utilize this application, special privileges have to be assigned in the Insign Server.

Users need one of the two privileges:

→ InSpire:monitorServer

→ InSpire:manageServer

Via this overview, the user can switch to individual workflows without any task and view a tabular and graphic display of the workflow history (i.e. what happened in the individual steps). The display is structured in 2 parts:

1. Graphic presentation of the workflow with the individual steps
2. Tabular presentation of the "events" relating to a workflow. The following information is found here:
 - a. Which user processed which tasks and at what time?
 - b. Which information was transferred to the WFM system (which variables were filled)?
 - c. Time of workflow instantiation

Selection criteria

The following selection criteria are available in the application:

Status

This selection criterion refers to the status of the workflow. Multiple selection is possible. The following statuses may be selected:

- Active
- Finished
- Finishing

- Invalid
- Invalid threads
- New

Please note: If no selection is made, selection is made according to the statuses Active and New by default, and only these workflows will be displayed.

Last modification

Selection according to the date on which the last modification was made. Warning: correct filtering is only possible if both the From and the To date are set.

Process name

Selection according to the process name. Please note: wildcard entry is not possible.

Workflow Overview Detail Application (Table)

In the tabular Workflow overview detail application, all workflows are displayed in accordance with the selection made.

The data available in the table are described below. These data might not be shown by default. In this case, they can be added using the column selection.

Process ID

Process ID

Process

Process

Version

Version

Status

Status of the workflow. Color display in accordance with the respective status.

Last modification

Last modification

Storage duration

Storage duration

Toolbar



Workflow History

Function authorization: wfhist

Link to function: Workflow history

**Workflow Information**

Function authorization: wfinfo

3 Workflow Events

Overview

Menu	System administration – Workflow management – Workflow events
Transaction code	wfev
Function authorization	wfev

Usage

The workflow events application shows instantiated workflows.

Prerequisite

Advanced object configuration

The workflow alias name must be defined in the "Advanced object configuration" application.

The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar. Inside the window, there's a 'Main page' header. Below it, there are two tabs: 'Default' and 'Last change'. The 'Default' tab is active. The form contains several fields: 'Object type' is a dropdown menu with 'WFM_CONFIG' selected; 'Object ID 1' is a dropdown menu with 'HYDRA_complaint_customer' selected; 'Object ID 2' is a dropdown menu with '1' selected; 'Object ID 3' and 'Object ID 4' are empty dropdown menus; 'Parameter' is a text field with 'Complaint_ComplaintType_KUNDE' entered; 'Parameter value' is an empty text field; there is a checked checkbox labeled 'Active'; 'Description' is a text field with 'Reference of customer complaint workflow template' entered; and 'Responsibility area' is an empty dropdown menu.

Object type

The fixed value WFM_CONFIG is to be defined here.

Object ID 1

The real workflow name is to be indicated here.

Object ID 2

The workflow version is to be indicated here.

Parameter

The workflow alias name is indicated here.

Selection criteria

The following selection criteria are available in the application:

Status

Status. The following statuses are available for selection:

- DONE
- DONE RESEND
- ERROR
- ERROR RESEND

Workflow Events Detail Application (Table)

The tabular workflow events detail application shows the workflows which were instantiated by HYDRA. If a workflow could not be instantiated, it can be instantiated again through this application.

The data available in the table are described below. These data might not be shown by default. In this case, they can be added using the column selection.

Workflow tab**Process ID**

Process ID

Status

Possible statuses:

- | | |
|--------------|--|
| DONE | → Workflow was instantiated |
| DONE RESEND | → Workflow was resent and then instantiated correctly |
| ERROR | → Workflow could not be instantiated |
| ERROR RESEND | → Workflow could again not be instantiated in a repeated attempt |

Status text

The error text of the workflow engine is shown here in the case of an error.

Module

Module which triggered the workflow.

Designation

Detailed designation of module

Process alias name

Process alias name

Process name

Process name

Process version

Process version

Workflow tab**ID 1-5**

ID 1-5 (references from HYDRA data base)

Key 1-5

Key 1-5 (text key from HYDRA data base)

Date key 1-5

Date key 1-5

Data

This shows the variables transferred, separated by pipe character.

Last editing tab**Editor**

Initially, this shows the editor who initiated the workflow. If the workflow is transferred again, the user is updated.

Last modification

Last modification

Trigger tab**Editor**

Editor who triggered the workflow.

Last modification

Time stamp of instantiation

Toolbar**Workflow History**

Function authorization: wfhist

Link to function: Workflow history

**Resend**

Workflows in ERROR or ERROR RESEND status can be transferred once again.

**Workflow Information**

Function authorization: wfinfo

4 Workflow Management - Task List

Overview

Menu	System administration – Workflow management - Task list
Transaction code	wftl
Function authorization	wftl

Usage

Using workflow management will generate tasks in a running process, which are to be handled by a user and/or a specified user. The task list shows the generated tasks for a user or a group assigned to the user. In addition, the user can process, forward and/or adopt a task.

The definition of the workflow and the related task definition are created in a separate designer in MES workflow management.

Selection criteria

The task list provides the following selection criteria:

User

By default, this field contains the user who is currently logged on. To change the user in order to display other tasks, the function authorization wftladmin has to be entered for the user currently logged on.

Show finished tasks

If this box is checked, finished tasks are also displayed. Otherwise only active tasks are shown.

Show the user's tasks only

If this box is checked, only the tasks directly assigned to the user are displayed. Otherwise the tasks assigned to a group to which the current user was assigned are also shown.

Created

Selection by the creation date of the tasks. Warning: correct limitation is only possible if both the From and the To date are set.

Done by

Selection by the date by which the task has to be finished. Warning: correct limitation is only possible if both the From and the To date are set.

Description

Selection by the task description. Wildcard search using * is possible here.

Task List Detail Application (Table)

The tabular Task List detail application shows all tasks matching the selections made.

The data available in the table are described below. These data might not be shown by default. In this case, they can be added using the column selection.

Process ID

Process ID of relevant task

Task ID

Task ID of relevant task

Work step

Name of work step. The name of the work step is automatically generated from the process element.

Role

The task may be assigned to a person, a group or a role. The individual assignments are displayed as follows:

USER: → Task is assigned to a person

GROUP → Task is assigned to a group

ROLE: → Task is assigned to a role

Title

The "Task description" of the generic task is displayed here.

Job

Job name of generic task

Description

Description of generic task

Priority

Task priority

The following values are possible:

Low, Normal, High, Urgent, Maximum

Created

Task creation date

Done by

"Done by" date of task

Finished

Termination time of task

Reserved for

Person for whom the task is reserved

Status

Status of task "not started", "reserved", "active" or "finished"

Thread ID

Thread ID

Flags

Flags

Send e-mail

Send e-mail

Total time

Total time

Estimated total time

Estimated total time

Locked by

Locked by

Started

Started

Calculation

Calculation

Classification

Classification

Costs

Costs

Deputy

Deputy

Url

Url

Resubmission

Resubmission

Last read

Last read

User data 1-4

User data 1-4

Toolbar**Process Task****Workflow History**

Function authorization: wfhist

Link to function: Workflow history

**Accept Task**

Acceptance of a task by the user currently logged on.

**Forward Task**

Forwarding of a task to a user, a group or a function (combination of group and role)

**Workflow Information**

Function authorization: wfinfo

5 Workflow History

Overview

Menu	-
Transaction code	wfhist
Function authorization	wfhist

Usage

The workflow history shows the sequence of a workflow in graphic and tabular form.

Selection criteria

The following selection criteria are available in the application:

Process ID

Process ID

Show details

The detail view shows all information relating to the workflow.

Workflow History Detail Application (Table)

The tabular Workflow history detail application shows the sequence of the workflow in accordance with the selections made.

The data available in the table are described below. These data might not be shown by default. In this case, they can be added using the column selection.

ID

ID

Thread

Thread

Step

Step

User

User

Time

Time

Parameter

Parameter

Previous value

Previous value

Status Detail Application (Image)

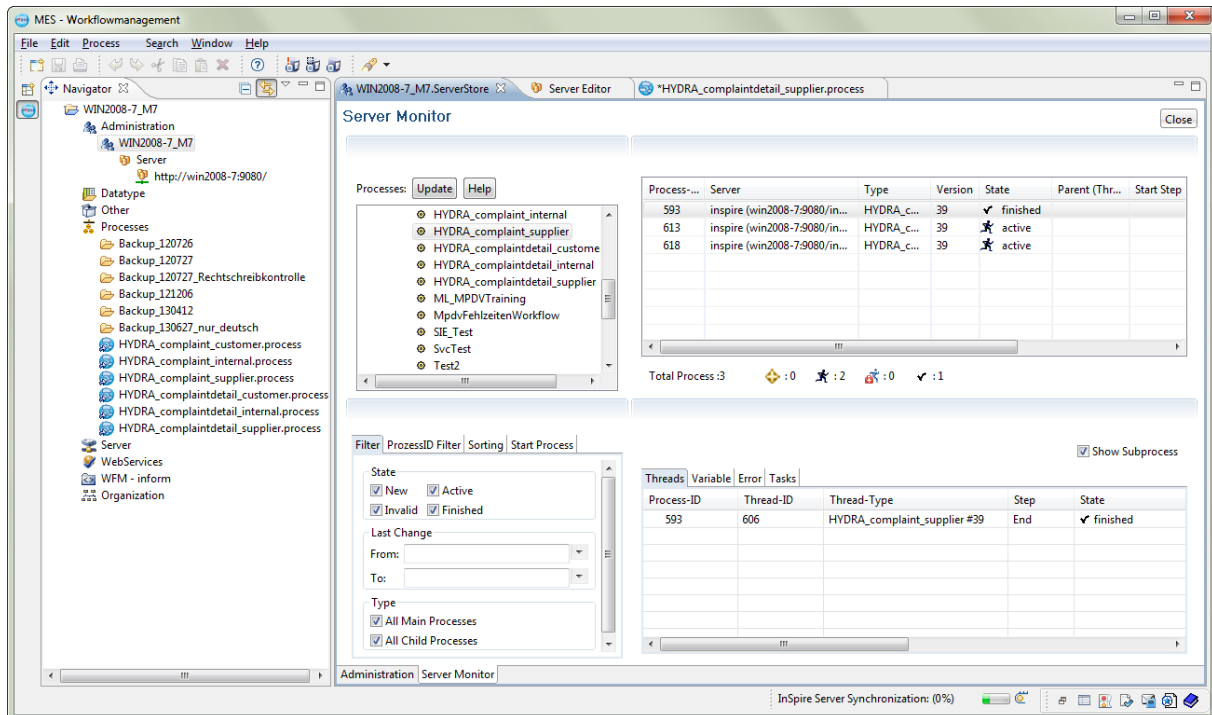
Zooming is performed by selecting the image and scrolling with the mouse wheel.

6 Workflow Designer

Overview

MES workflow management enables the mapping of processes by means of a workflow. The workflows used for this are instantiated and thus started by HYDRA. In order for a workflow to be used, it must be created in a previous step in the workflow designer and assigned with appropriate parameters.

Workflow designer screenshot:



The workflow designer is not integrated in MOC and is implemented on the basis of Eclipse. In addition to the administration of the workflow management server, it also provides the interface for creating and deploying workflows.

In the initially installed standard status, the components required for using the workflow management and the workflow management database are found on the HYDRA server. This can be opened here via a link in the HYDRA administration file and/or via the Start menu by using the MES workflow management - designer link.

Prerequisite

JAVA 6 must be installed on the PC where the MES workflow designer is to be used.

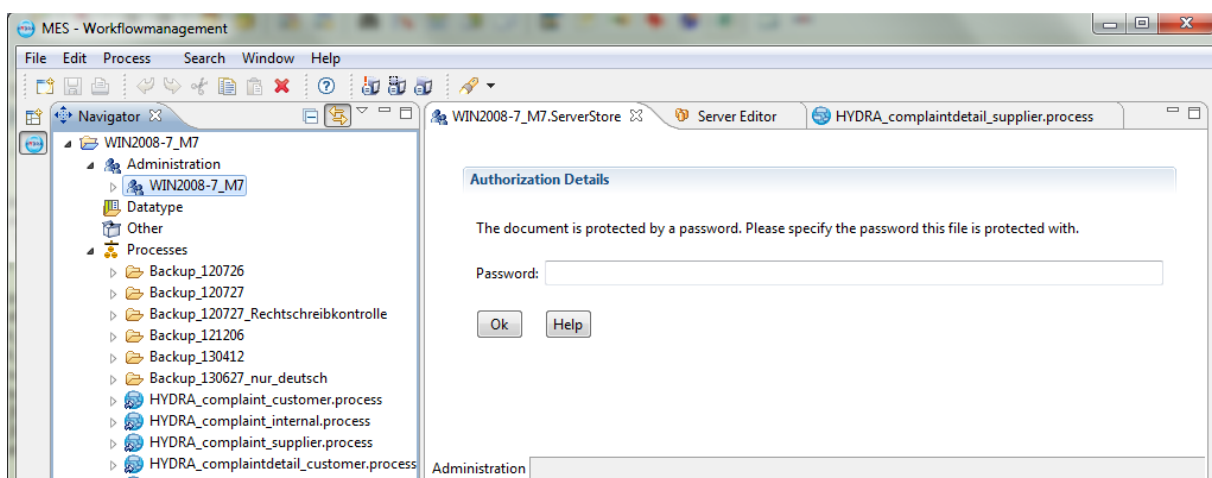
The server-side installation of workflow management must have been performed by MPDV commissioning. The server address and the workflow management port must be known. The password of the workflow administration user (e.g. admin) must be known.

The user must be familiar with the general characteristics of the workflow designer:

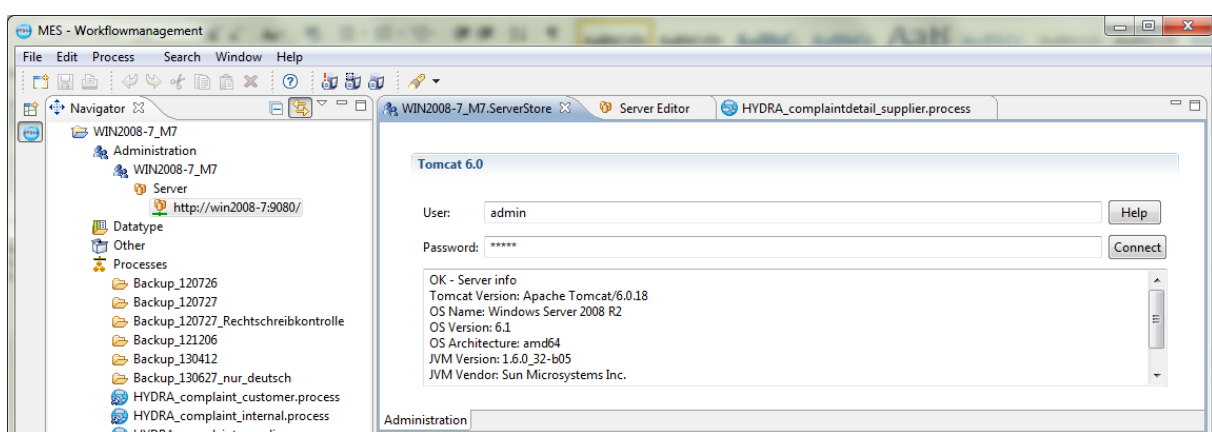
- Process server monitoring
- Process design
- Process deployment

6.1 Starting the Workflow Designer

When opening the workflow designer and double-clicking the Tomcat menu item (or the assigned name), the password assigned for the administration document must be entered and subsequently be confirmed by pressing the OK button so that a link to the workflow management server can be established.



Upon successful entry and connection to the workflow management server, the following information is displayed:



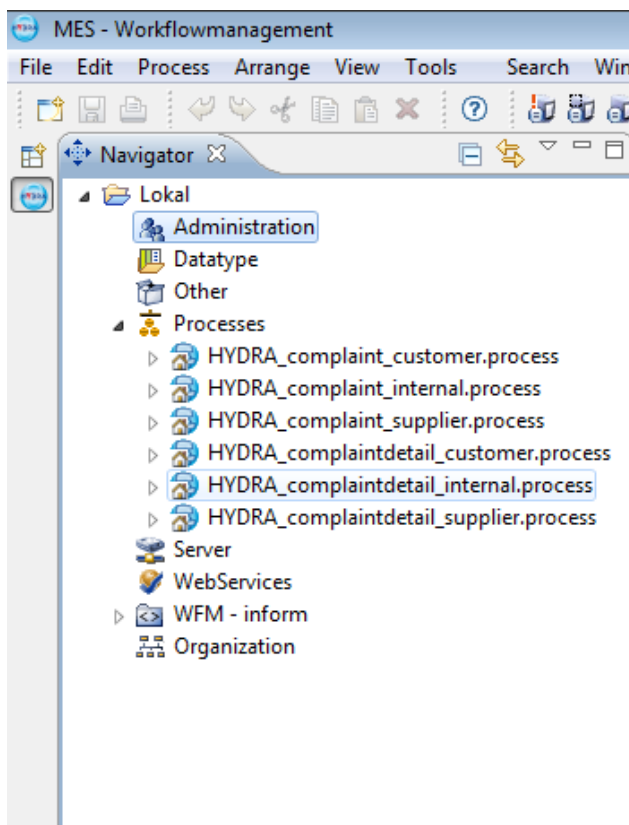
Once the connection has been made successfully, the following sub-items are found under the server entry (in this example under the entry: <http://win2008-7:9080/>):

- WFM – Admin Server
The sub-items in the WFM – Admin Server include the master data and/or settings required for the process of the workflows. The master data such as the user, however, are retrieved from HYDRA and synchronized with the workflow management system.
- WFM – Processes Server: inspire
The processes server includes all deployed workflows and the associated required parameters.

6.2 Workflows

Creation of a New Workflow

By double-clicking the Processes item (see screenshot) and/or a folder created in it, the entry screen for the name of the new process opens. The naming convention is to be observed (please refer to section 6.8).



After entering the name, the new process is shown in the list of processes and, once closed, can also be re-opened by double-clicking the process name.

Design Guidelines for HYDRA Workflows

The following general conditions must be observed when designing a HYDRA workflow:

- The workflow must have precisely one starting point.
- The workflow must have at least one ending point. As an option, several ending points may be used.
- The workflow should, if possible, not call any sub-processes since these cannot be displayed in the workflow overview in MOC.

In the following sections, it is noted for several description fields that these serve 'technical documentation'.

The contents of these fields will not be visible for the editor of workflow tasks. For the designer of a workflow, however, these descriptions provide a quick overview as to what is hidden behind the variables, the e-mail templates and the work steps.

For this reason, this information should always be provided, since it significantly facilitates support and further development.

Workflow design

In addition to other options, the drawing surface for the workflow can be opened at any time by double-clicking on the workflow created. The following options for designing a workflow are available here:

Symbol	Designation
	Selection
	Shift view
	Zoom
	Text
	Positioning a pool
	Positioning the starting point of a process
	Positioning an ending point of a process
	Positioning a process step
	Creation of a sub-process call
	Creation of a multiple decision
	Positioning a decision element
	Positioning a 'Go to' / 'Merge' element



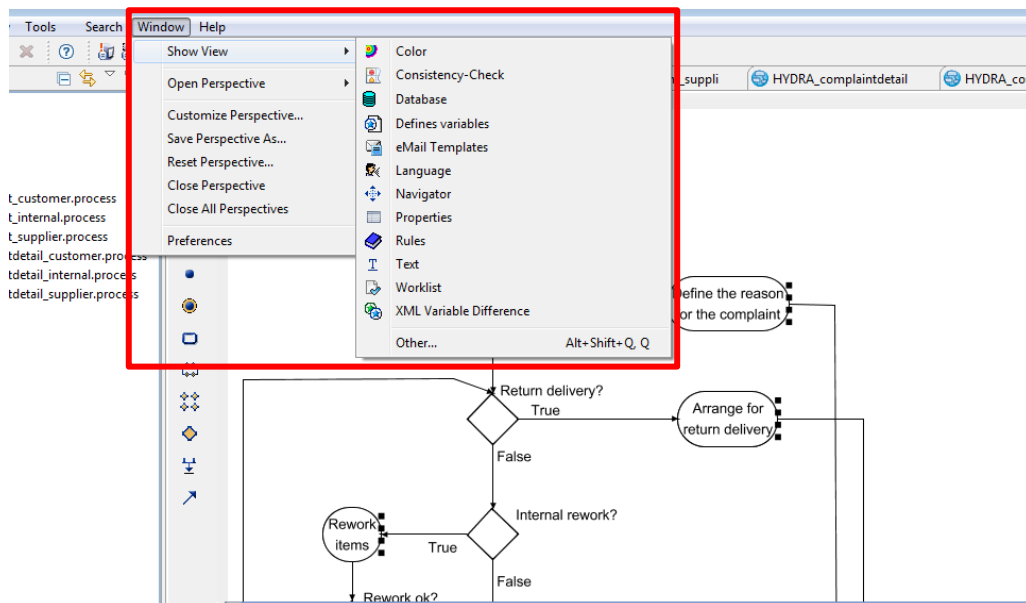
By drag & drop, the individual elements may be transferred to the drawing area for creating a workflow.

Properties for a Workflow

The following information can be maintained and functions called up for a workflow:

- **Properties**
The properties of the currently highlighted workflow are displayed here
- **Consistency check**
A current display of potential warnings, errors and information on the complete workflow is shown here
- **Working list**
This includes the defined activities to be performed for the currently highlighted workflow step.
- **Define variables**
In order for variables to be used in the individual actions of the workflow steps, they must first be defined and entered in the workflow. Please note: The naming conventions for variables are to be observed.
- **E-mail template**
If an e-mail is to be sent using a defined e-mail template in an action, this template must first be defined on the workflow.
- **Rules**
The defined rules relating to decisions in the workflow are maintained here. Only the rules defined in the current workflow step are displayed.

If one of the views specified is not available in the workflow designer, the required views can be displayed via the menu item Window → Show view (see screenshot).



Entering Variables

For using variables in the entire workflow, regardless of whether they are used in e-mail templates or for generic tasks, they have to be entered in the "Define variables" view on the workflow.

Here, variable names, the associated data type, a default value and a language-dependent description may be entered. For allocating variable names, the naming conventions are to be observed (see section 6.8).

In addition, a description may be entered. This is used for technical documentation and can be entered in several languages.

If a task screen in MOC is to be defined by means of the Java Inline Code, the fixed variable `taskData` must first be defined in the variables on a workflow ("Long string" data type).



Should parallel processes occur in a workflow, the related process step-specific variables `<Name_des_Prozessschritts>_taskData` must be defined.

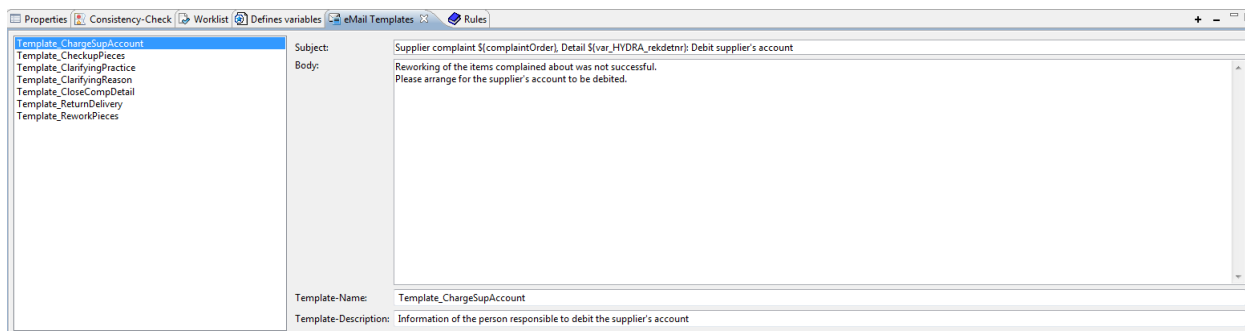
Entering E-Mail Templates

To be able to use e-mail templates in various actions, they must have been entered on the workflow in a first step. They can then be used and/or allocated in the individual actions via the template name.

When the e-mail is sent, the information in the header line is shown as the Subject and the information in the Mail Text cell is used as the e-mail text.

In addition to fixed texts, variables and line breaks can be used with the customary HTML formats for structuring, e.g. `<br>` for a line break.

Variables can be used in the entire e-mail template by means of the syntax `${var_VariablenName}`.



When defining e-mail templates, the naming conventions are to be observed (see section 6.8).

The header and the e-mail text of the template can be entered in several languages.

In addition, a description may be entered. This is used for technical documentation and can be entered in several languages.

6.3 Elements of a Workflow Design

Starting Point

In the starting point properties, the Description parameter can be multilingual.

Ending Points

In the ending points properties, the Description parameter can be multilingual.

Process Steps

For some added process steps and/or decisions, the required parameters may be entered using the displayed views.

If a required view is missing, it can be displayed as described above.

In the process steps properties, the Description parameter can be multilingual.

Decision Elements

In the decision elements properties, the Description parameter can be multilingual.

Multiple Decisions

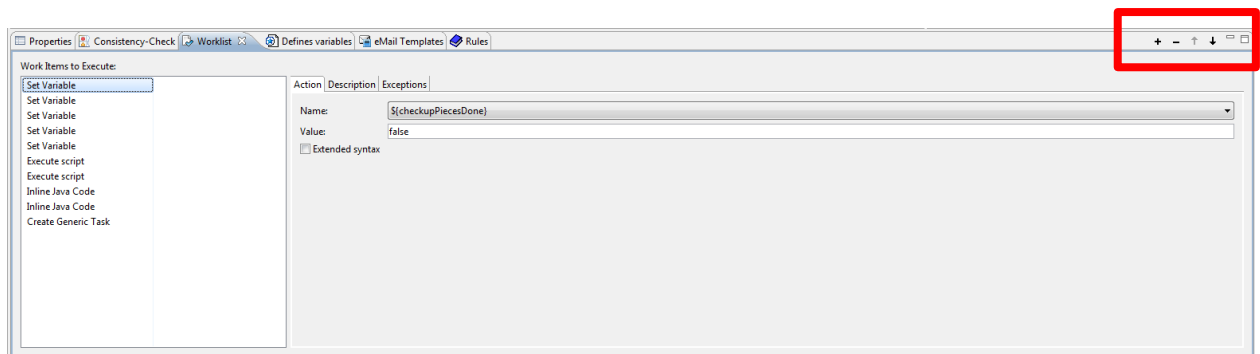
In the multiple decision properties, the Description parameter can be multilingual.

Links

In the links properties, the Description, Labeling 1 and Labeling 2 parameters can be multilingual.

6.4 Definition of the Working List of a Process Step

Using the working list context menu and/or the relevant buttons (see red marking), individual actions may be assigned to the process steps. Actions may be assigned several times or only once, depending on the action type.



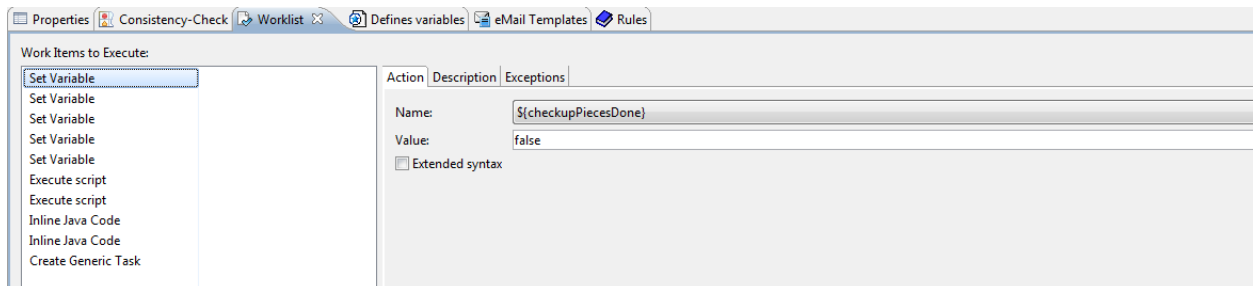
The following actions may be chosen:

- Set variable value
- Send e-mail
- Set xml variant
- Assign XML structure
- Inline Java Code
- Rule only
- Variable -> XML
- Create generic task
- Re-start workflow
- Start workflow
- Terminate workflow
- Call web service
- Execute simple EAI adaptor
- Execute script

For detailed information, please refer to the designer's Help function, which you can, for example, open directly via the Start menu under the Documentations item.

Set variable value

This action may be used to assign values to the variables entered in the workflow. The variable to be set may be explicitly set with a value entered in the Value field using the drop-down box. All variables which were entered on the current workflow in the "Define variables" view are available (see section 6.2 - Entering variables).

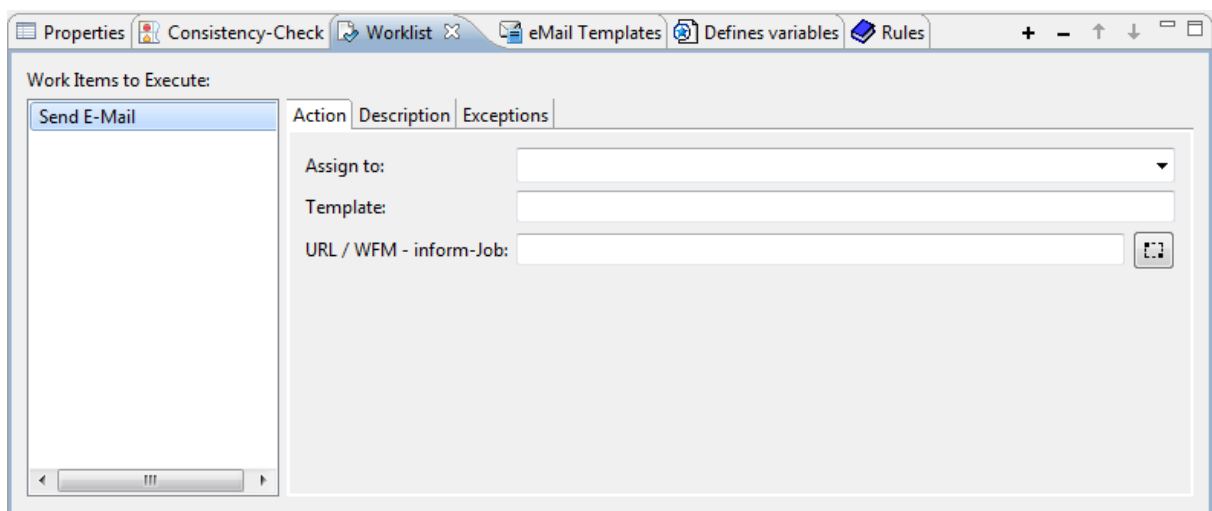


In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Send e-mail

Through the "Send e-mail" action, an e-mail can be sent to a user and/or a group or function by entering an e-mail template.

All e-mail templates available are defined in the "E-mail templates" view on the workflow and can be sent by assigning the name in a process step.



In the "Assign to" field, fixed values such as User abcd, but also variables may be used, which may include various addressees according to the setting of the variable in the process.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Please note: The technical prerequisites for sending e-mails are already met upon system startup.

Set xml variant

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

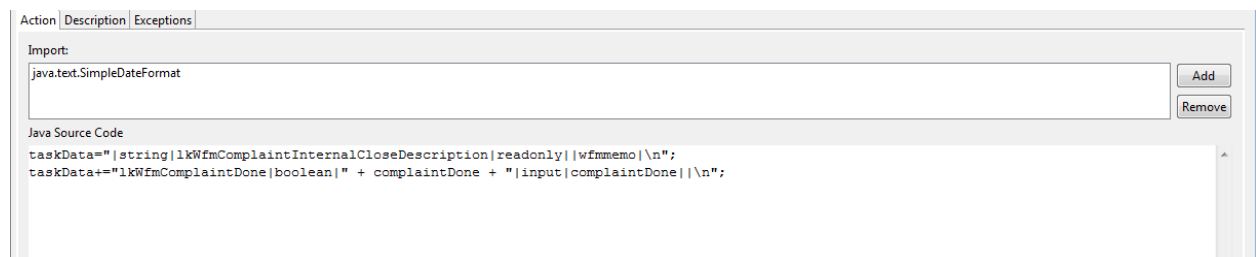
Assign XML structure

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Inline Java Code

The "Inline Java Code" action may be used to program functions by means of JAVA.



In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

By means of JAVA programming, for example, a description of the "Edit task" screen in MOC can be defined e.g. by supplying the variable `taskData`. This determines the appearance of the task screen here in the designer. The technical details of this solution are described below:

Prerequisite: The `taskData` variable must be added to the process with the "Long string" data type (see section 6.2 - Entering Variables).

Any field to be displayed in the task processing later is assigned in a separate line in this example for the `taskData` variable. The individual parameters which configure the field are to be separated by a pipe. The parameter set of a field is ended with `\n`.

Example for an Inline Java Code line:

```
taskData="Fertig?|boolean|" + taskDone + "|input|taskDone||\n";
```

The individual configuration parameters of a field defined by the `taskData` variable have the following meaning. The position here describes where the parameter is located in the pipe-separated list.

Pos.	Parameter	Description
1	Field label	This parameter determines the label in the MOC editing screen. If MOC task editing is to be available in one target language only, the label can be directly indicated here. Example: <pre>taskData="Fertig? boolean " + done + " input done \n"</pre> If task editing, however, is to be used in several languages, an entry in the MOC dictionary can be referenced by a corresponding language key at this point. Example: <pre>taskData="lkWfmDone boolean " + done + " input done \n"</pre>
2	Field data type	The field data type also determines the presentation in MOC. It must comply with the type of the associated variable (see parameter 5 and 3, where applicable). Example: <pre>taskData="Palette string " + palId + " input palId \n"</pre> The following data types are available here: boolean This data type represents a logic value which may assume the values <code>true</code> or <code>false</code>. In MOC, the field is displayed as a checkbox. string This data type represents a character string. In MOC, the field is displayed as an input field. datetime This data type represents a point in time (comprising date and time). In MOC, the field is displayed as an input field for times.

3	Value to be displayed	Value to be displayed in the field upon opening the editing dialog in MOC. At this point, a constant parameter may be entered. Example: <pre>taskData="Palette string P345 input palId \n"</pre> If this parameter is omitted completely, this can achieve the effect that the field is empty when opened. Alternatively, the contents of a defined variable may be displayed. Example: <pre>taskData="Palette string " + palId + " input palId \n"</pre>
4	Editing/Display status	If the MOC user should be able to edit the field, <code>input</code> is used here. Example: <pre>taskData="Palette string input palId \n"</pre> Optionally, <code>readonly</code> is displayed. Example: <pre>taskData="Palette string readonly palId \n"</pre>
5	Input target	In the variable defined here, the contents of the associated field are saved when the task is saved in MOC. Example: <pre>taskData="Palette string " + palId + " input palId \n"</pre>
6	Syntactic type	Syntactic type which defines how the value is to be displayed in MOC. Example: <pre>taskData="Karte string " + kNr + " input kNr user_id \n"</pre> By using the syntactic type <code>wfmmemo</code> you can achieve the effect that text information (<code>string</code> data type) is shown on several lines. Example: <pre>taskData=" string " + iTxt + " readonly wfmmemo \n"</pre> For entries of the syntactic type <code>wfmmemo</code>, an entry in the MOC dictionary can also be referenced via a corresponding language key instead of 'fixed contents'. This allows for optionally designing the contents of this multi-line field in several languages (e.g. in order to provide the user with notes on task handling). Example: <pre>taskData=" string lkWfmTaskDone readonly wfmmemo \n"</pre>

- | | | |
|---|----------|--|
| - | Line end | At the end of the parameters of a field, the line end \n is always required. |
|---|----------|--|

The first field to be indicated in the MOC editing screen of the task is assigned to the `taskData` variable as follows in this example:

```
taskData=<Parameter of first field>
```

If further fields are to be added, the relevant definitions may be added to the `taskData` variable as described below:

```
taskData+=<Parameter of second field>
taskData+=<Parameter of third field>
:
```

Using the `infoData` variable, a workflow detail dialog can be started, which will display fields for the entire workflow regardless of the current task. The field contents may also be edited in MOC as an option.

When configuring the fields of this dialog, the same functions as described above are applicable: However, for this dialog, the `infoData` variable instead of `taskData` is supplied with the field configurations.

When using **tasks in parallel**, the definition of fields for maintaining tasks by means of one variable would cause problems.

In order to avoid this, the definition of field configurations is not assigned to the `taskData` variable in this case. Instead, the name for this variable is formed as follows:

```
<Name_of_related_process_step>_taskData
```

The name of the related process step can be taken from the properties of the process step.

Rule only

The "Rule only" action may be used to define an end condition for a process step behind one and/or several process steps.

The "Rule only" action can only be entered once at a process step. If an end condition has already been defined for a process step in another action, the "Rule only" action cannot be added any more.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Variable -> XML

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Create Generic Task

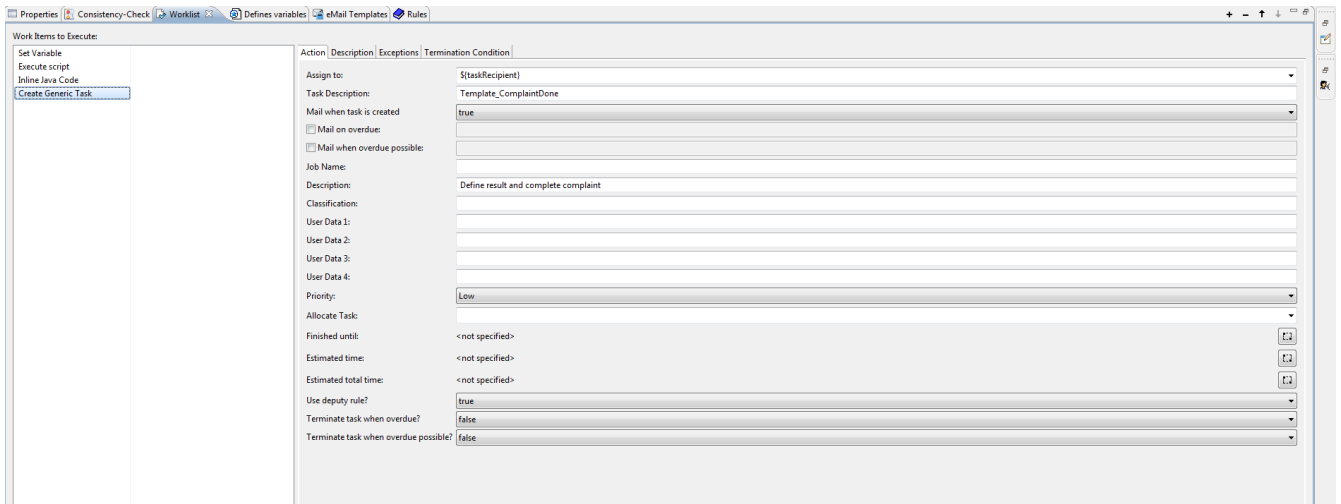
The "Create Generic Task" action can be used to enter tasks on the workflow steps, which must be performed before the workflow can be continued. Example: A responsible person shall decide whether or not a complaint will be processed.

For this purpose, the action for a generic task and the Java Inline Code action for the task screen in MOC must be entered.

In the Action tab, the following specifications may be defined for a generic task:

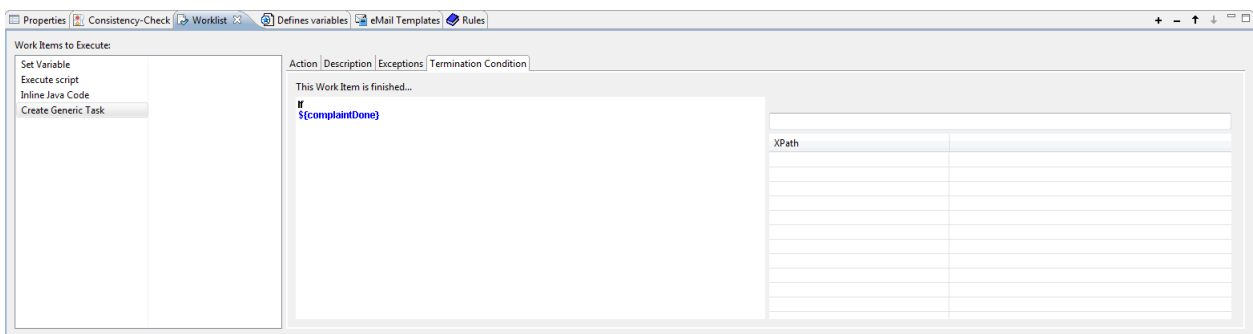
- **Assign to:**
This field will assign the task, e.g. which user is assigned with the task. In addition to fixed values, variables may also be assigned, which are for instance determined in a previous action and/or in an upstream process step.
Depending on whether a task is assigned to a USER, a GROUP or a FUNCTION, the type and the related name have to be defined e.g. by a variable.
- **Task description**
A brief description of the task may be entered here in several languages. If an e-mail is to be sent (see next item), the task description is used as the name for the e-mail template.
- **E-mail when creating the task**
Definition of whether an e-mail is to be sent when the task is created. If so, an e-mail template using the name of the task description is sought.
- **E-mail if due date is exceeded**
- **E-mail if due date might be exceeded**
- **Job Name**
- **Description**
Multilingual field for task description
- **Classification: <Future use>**
This field can be multilingual.
- **User data 1-4: User-specific data may be entered in these fields.**
- **Priority**
The priority is shown in the task list on MOC and sorting/grouping is possible.
- **Reserve task: <Future use>**

- Done by
- Estimated duration
- Estimated total duration
- Apply absence management
Shall defined absence management be applied?
- Terminate task if due date is exceeded?
- Terminate task if due date might be exceeded?



In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

A terminate condition for this process step may be defined under the "End condition" tab. In this example (see Screenshot), the End condition is reached if the `$(var_ok)` variable is assigned with `true`. To prevent that this is the case from the start already, the value may explicitly be set to `false` by means of the "Set variable value".



Re-start workflow

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Start workflow

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Terminate workflow

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Call web service

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

The description in the tab with the same name can be multilingual.

Execute simple EAI adaptor

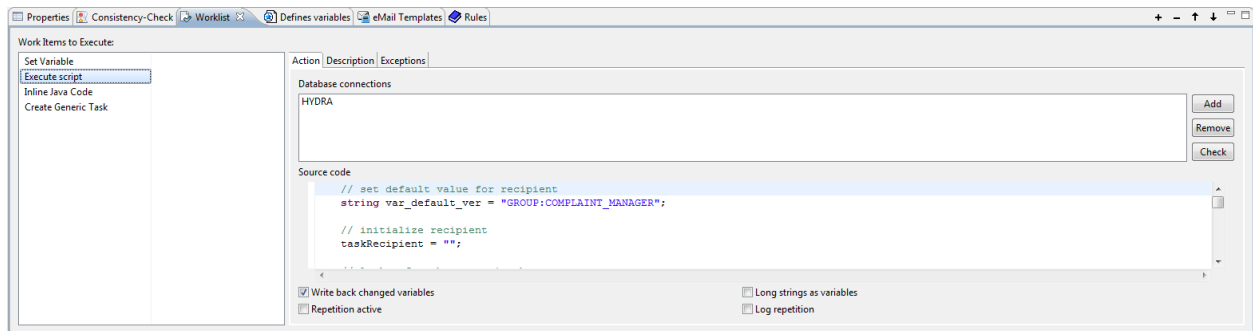
Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Execute script

In the current process, HYDRA data saved in the database may be accessed from the workflow. For this purpose, the "Execute script" action type must be used. The HYDRA database is accessed via JDBC.

A database link must be added first. The Add button opens a list with the configured databases. In the following example, the database link "HYDRA" was selected.



In this example, other machine master data (designation and cost center) are read. The SQL statement is assigned to the `sel` string. At this point, the SQL keyword "select" is not indicated!

```
string sel = "bez_lang, kostenstelle from maschinen where masch_nr = '" + var_mnr + "'";
// Mit set_first kann nach einem erfolgreichen Select geprüft werden,
// ob Daten gefunden wurden.
if (select("mnr", sel, "HYDRA") && set_first("mnr"))
{
    // Schleife über die Zeilen.
    //do {
        var_bezl = get_value("mnr",0);
        var_kst = get_value("mnr",1);
    //} while (set_next("mnr"));
}
// Datenobjekt freigeben
end_select("mnr");
return true;
```

The do - while loop was omitted since only one data record is expected in the result data quantity. The machine designation and the cost center are assigned to the relevant process variables.

The "Apply modified variables" option must be selected.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

6.5 Deploying Workflows

After completion of a workflow, it must be deployed via the workflow designer so that it is distributed on the server and can thus be executed. The relevant buttons are found in the toolbar:

- 

Quick Deploy of modified ME workflow management resources – deployment is triggered for all modified workflows without any selection or confirmation
- 

Quick Deploy of selected MES workflow management resources – after selecting the workflows to be deployed, deployment is triggered for the selected workflow.



Deployment of MES workflow management resources – after selecting the project and the workflows to be deployed, deployment is triggered for the selected workflows.

If the workflow to be deployed contains errors, deployment is not executed and the errors are listed.

Whether deployment is executed with existing warnings can be selected when deploying.

After deployment, the workflows are available on the server and in the workflow designer under the item WFM Process Designer: inspire → Process Definitions, and processes may be started and/or current processes may be viewed for the different versions.

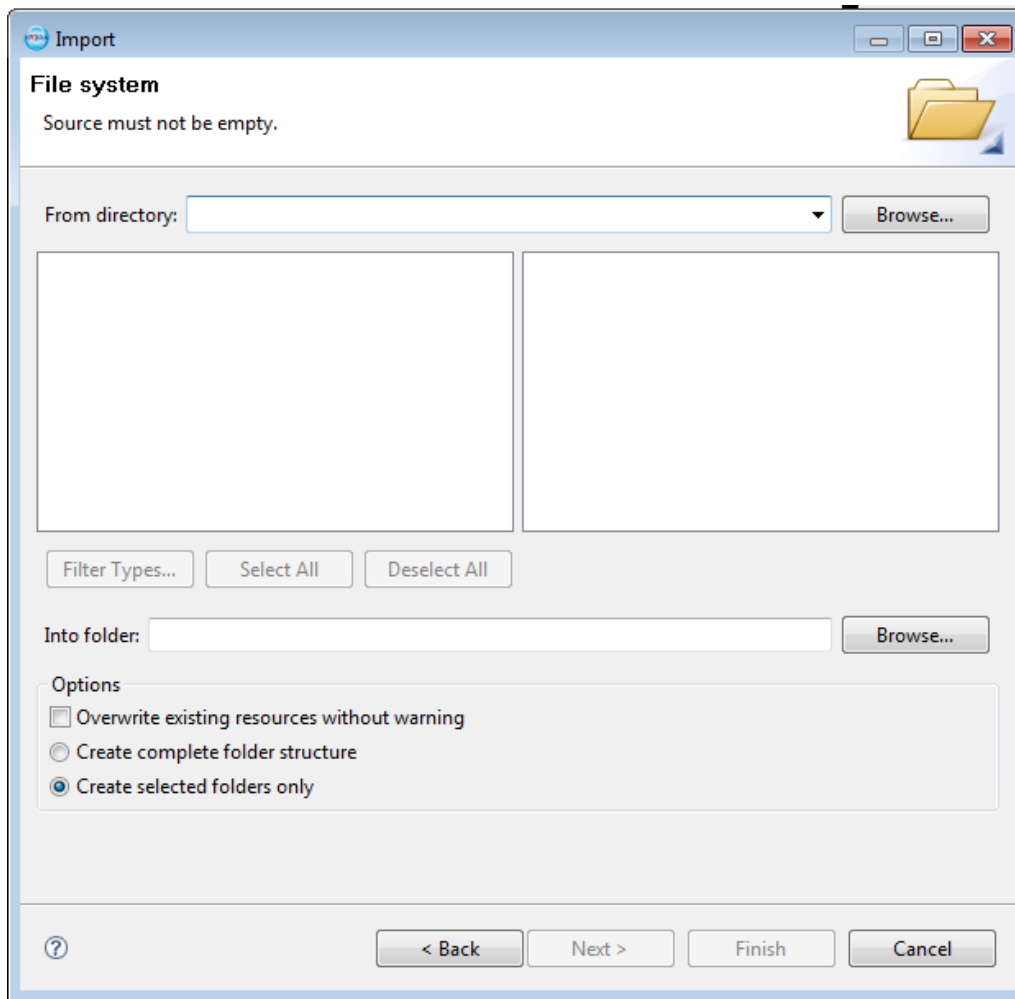
6.6 Import/Export of Workflows

Import of Workflows

In order for existing workflows to be edited in the workflow designer, these have to be imported first.

Under the File → Import to menu item, existing and exported workflows can be imported.

Screenshot:



The default workflows which can be used as default for system startup can be found under the following directory on the HYDRA server:

<\\<HYDRA-Server>\<HYDRA-Verzeichnis>\products\wfm\>

Export of Workflows

In order to save and/or distribute existing workflows, these may be exported from the workflow designer and filed on the target system by means of the File → Export to menu item.

6.7 Language / Multilingual input

The Language view can be used to enter designations in several languages in individual fields, e.g. a workflow step designation.

For this purpose, the field where the individual languages are required has to be maintained appropriately. If a language is not maintained, the default language will be displayed in the workflow designer.

Please note: The defined default language for the created workflows is German.

6.8 Naming Conventions

Naming Conventions for Workflows

The names of process steps and links must be English and follow the conventions described below:

- Start:
Start:
- End:
End or
End_<Description in PascalCase>
Example: End_Success
- Process steps:
Step_<Description in PascalCase>
Example: Step_InvestigateTask
- Decisions / Multiple decisions:
Decision_<Description in PascalCase>
Example: Decision_Justified
- Links:
Link_<Description in CamelCase>
Example: Link_DoReworkCustomer
Link_ReworkCustomerDone
Link_ArrivalConfirmed
- Splitting:
Split_<Description in CamelCase>
Example: Split_UrgentCause
- Merger:
Merge_<Description in CamelCase>
Example: Merge_8dClose

Character string variables whose contents will affect workflow control (e.g. simple multiple decision using precisely one variable) should have English contents, if possible.

The labels used for the associated elements in the taskData variable must be English and follow the conventions below:

- lkWfm_<Description in PascalCase>
Example: lkWfmComplaintDataCollected

If the name of the variable was selected appropriately, there is nothing wrong with setting the variable name behind the prefix lkWfm. If a generic task was assigned to a process step, the description of this task can usually correspond to the text of the process step in the sequence chart. The description of a generic task usually matches the description of the associated process step.

Naming conventions for e-mail templates

The header of the e-mail template of a generic task usually includes the description of the generic task. As a prefix, the respective context should also be included. The label for entering a variable (in the context of a task) in HYDRA should usually match the description of the relevant task.

The names of the e-mail templates must be English and follow the conventions described below:

- Template_<Description in PascalCase>
Example: Template_CollectData

Naming Conventions for Variables

The names of the variables must be English and follow the conventions described below:

- General variables
<Description in CamelCase> Example: `complaintDataCollected`
- Variables whose contents are provided by HYDRA upon instantiation of a workflow
`var_HYDRA_<Name of field in HYDRA in lower case>` or
`var_HYDRA_<Name of parameter in HYDRA in PascalCase>`
Example.: `var_HYDRA_reknr`
`var_HYDRA_InspectionStep`

6.9 Documentation of Created Workflows

For the documentation of edited and/or created workflows, 2 documentation types are distinguished:

- User-related documentation on workflow
- Technical workflow documentation

User-related Documentation on Workflow

User-related documentation should primarily contain the defined workflows with individual workflow steps from the end user's perspective. It must be ensured that a graphic representation of the entire process on the one hand, but also a definition of the individual process steps with the underlying actions and/or tasks on the other hand, are provided for the end user.

Technical Workflow Documentation

Via the Process → Generate document menu item, technical documentation can and/or must be prepared for the created workflows. The contents to be used for documentation may be selected in individual steps. The following contents are available here:

- Variables

- E-mail templates
- Merger
- Decision
- Sub-process
- Selection
- Step
 - Step
 - Working list
 - Action
 - Exception

Documentation should be prepared as rtf, since in addition to the text components a screenshot of the designed workflow will subsequently have to be integrated in the documentation

7 Workflow Management Configuration

Overview

In order to utilize workflow management, settings in HYDRA have to be made via MOC in addition to installing the required WKF components.

Prerequisite

JAVA 6 has to be installed on the PC where the MES workflow designer is to be used.

The commissioning staff of MPDV must have installed workflow management on the server. The server address and the port of workflow management must be known. The password of the workflow administration user (e.g. admin) must be known.

Advanced Object Configuration

When a workflow is instantiated, the defined and required data are sent to workflow management and a workflow is started there. The data required include those identifying which process (workflow) is to be started in which version.

The definition as to which workflows are available and which is the version to be activated is performed via the advanced object configuration (adoc) maintained in HYDRA. The related entry has the following appearance:

Object type	Object-ID 1	Object-ID 2	Object-ID 3	Object-ID 4	Parameter	Parameter value	Active
WFM_CONFIG	<Workflowname>	<Workflowversion>	-	-	<abstr. Workflowname>	Irrelevant	Y

Please note:

- If no entry is found, the abstract name and zero as version are returned.
- If no version was defined for a workflow, the last version deployed is always used in the workflow management system.

User configuration

For entering users in the workflows, the defined users are transferred from HYDRA to Inspire by manually triggering the synchronization and through the cyclical scheduler job.

Important: In order to be able to recognize whether users were imported from HYDRA or entered directly in Inspire, users imported from HYDRA are assigned the attribute "Origin_HYDRA=true".

The following updates are run for users during synchronization:

- HYDRA users missing in Inspire are created in Inspire with the parameters defined and with the attribute "Origin_HYDRA".
- Existing HYDRA users with the "Origin_HYDRA" attribute are updated.
- Inspire users bearing the Origin_HYDRA attribute and not/no longer existing in HYDRA are removed.

At present, the following parameters are transferred from HYDRA to Inspire for users during synchronization:

- User name
- First and last name
- E-mail address
- Privileges

Language Settings on User

Apart from the parameters transferred from HYDRA, the user has additional attributes in Inspire, for example, the InSign:USER_LOCALE attribute, in which the user language of the user is defined. This means that when the user requests his/her tasks created by workflow management from HYDRA, the language defined for the user in Inspire is used for the task display.

If the language setting is to be changed, the InSign:USER_LOCALE attribute has to be modified manually for the relevant user in the workflow management system (Inspire). The language settings in MOC are not used for this.

Synchronization of Privileges

Privileges are synchronized last of all. In this process, all function authorizations starting with wfm:... as defined in HYDRA as well as the authorizations from function profiles starting with wfm:... are defined as privileges in Inspire, and authorizations existing in Inspire but not in HYDRA are removed (only for users with Origin HYDRA = true).

The mapping between function authorizations and InSign / Inspire privileges is as follows:

InSign.adminUser	wfm:a-au
InSign.logonAnyUser	wfm:a-lau

InSign.manageAnyObject	wfm:a-mao
InSign.manageApplication	wfm:a-ma
InSign.manageClients	wfm:a-mc
InSign.manageServer	wfm:a-ms
InSign.readAnyCalendar	wfm:a-rac
InSign.serverUser	wfm:a-su
InSign.writeAnyCalendar	wfm:a-wac
InSpire.adminUser	wfm:p-au
InSpire.deploy	wfm:p-d
InSpire.goToAnyProcess	wfm:p-gtap
InSpire.manageApplication	wfm:p-ma
InSpire.manageObjectPrivileges	wfm:p-mop
InSpire.manageObjectPrivileges.inspire	wfm:p-mopi
InSpire.manageObjects	wfm:p-mo
InSpire.manageObjects.inspire	wfm:p-moi
InSpire.manageServer	wfm:p-ms
InSpire.monitorServer	wfm:p-mons
InSpire.readAnyProcess	wfm:p-rap
InSpire.readAnyTask	wfm:p-rat
InSpire.serverUser	wfm:p-su
InSpire.start	wfm:p-s
InSpire.terminateAnyProcess	wfm:p-tap
InSpire.writeAnyProcess	wfm:p-wap

InSpire.writeAnyTask	wfm:p-wat
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